Nova Analytics - EDA Report

Dataset: Fish.csv

159 rows x 7 columns

Dataset Information

Shape

Rows	**159**
Columns	**7**

Data Types

Column	Data Type	Non-Null	Null	Unique
Species	object	159	0	7
Weight	float64	159	0	101
Length1	float64	159	0	116
Length2	float64	159	0	93
Length3	float64	159	0	124
Height	float64	159	0	154
Width	float64	159	0	152

Descriptive Statistics

Column	Mean	Median	Std Dev	Min	25%	50%	75%	Max
Weight	398.33	273.00	357.98	0.00	120.0	273.0	650.0	1650.00
Length1	26.25	25.20	10.00	7.50	19.05	25.2	32.7	59.00
Length2	28.42	27.30	10.72	8.40	21.0	27.3	35.5	63.40
Length3	31.23	29.40	11.61	8.80	23.15	29.4	39.65	68.00
Height	8.97	7.79	4.29	1.73	5.94	7.79	12.37	18.96
Width	4.42	4.25	1.69	1.05	3.39	4.25	5.58	8.14

Missing Value Analysis

Total Missing Values: **0**

Column	Missing Count	Percentage
Species	0	0.0%
Weight	0	0.0%
Length1	0	0.0%
Length2	0	0.0%
Length3	0	0.0%
Height	0	0.0%
Width	0	0.0%

No missing values detected in this dataset.

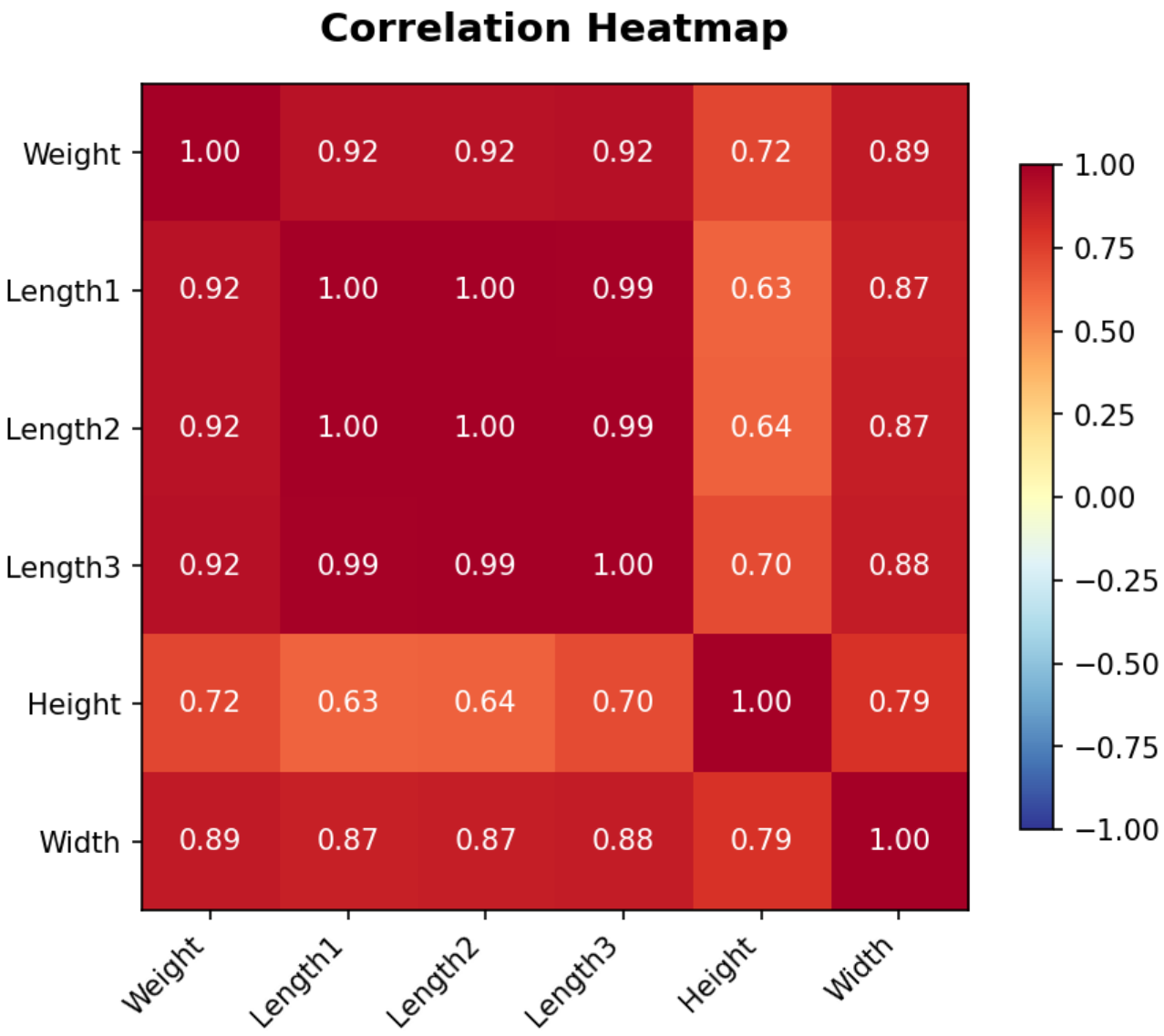
Duplicate Analysis

Duplicate Rows	**0 (0.0%)**
Total Rows	**159**

Correlation Analysis

Correlation Matrix

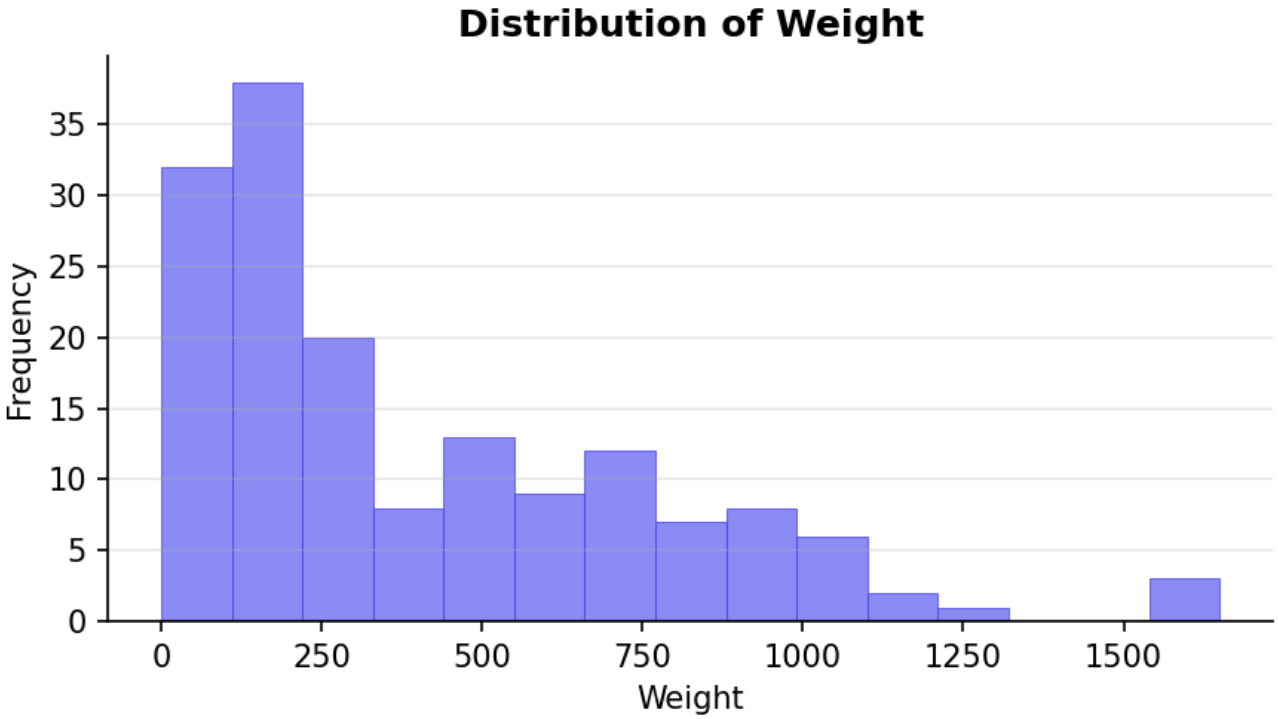
	Weight	Length1	Length2	Length3	Height	Width
Weight	1.00	0.92	0.92	0.92	0.72	0.89
Length1	0.92	1.00	1.00	0.99	0.63	0.87
Length2	0.92	1.00	1.00	0.99	0.64	0.87
Length3	0.92	0.99	0.99	1.00	0.70	0.88
Height	0.72	0.63	0.64	0.70	1.00	0.79
Width	0.89	0.87	0.87	0.88	0.79	1.00

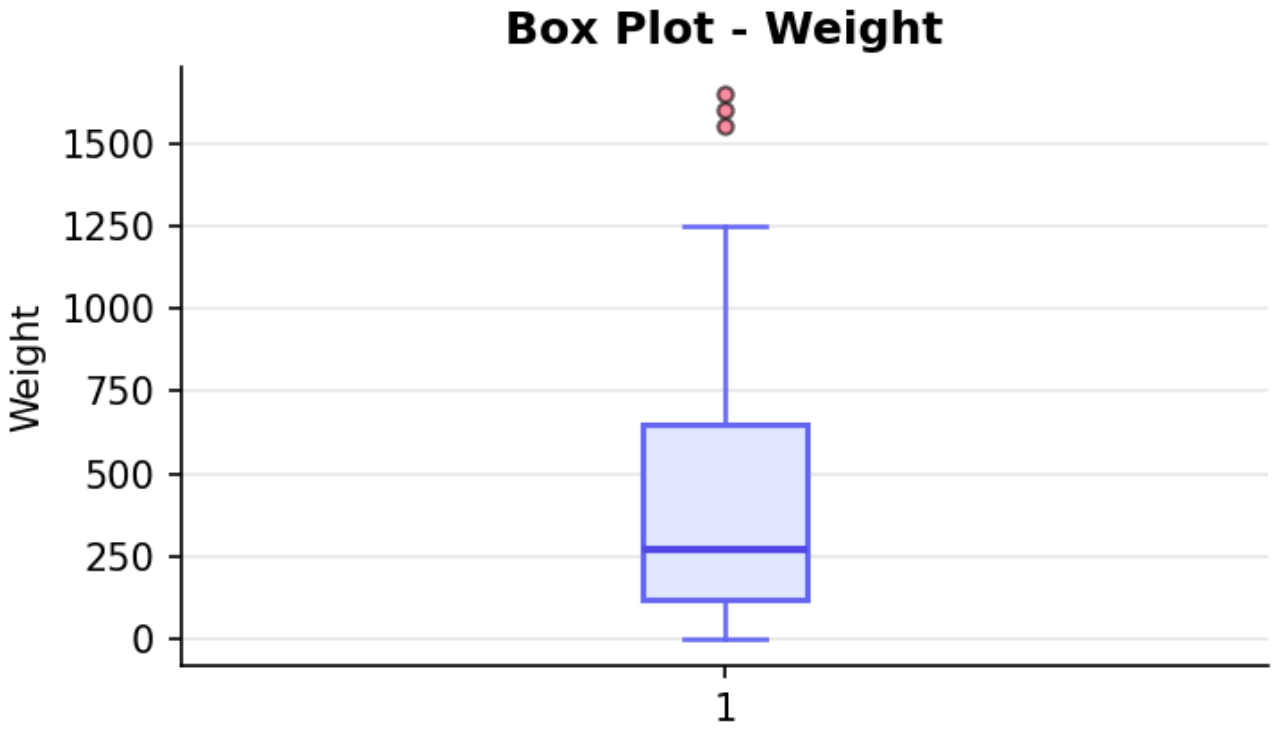


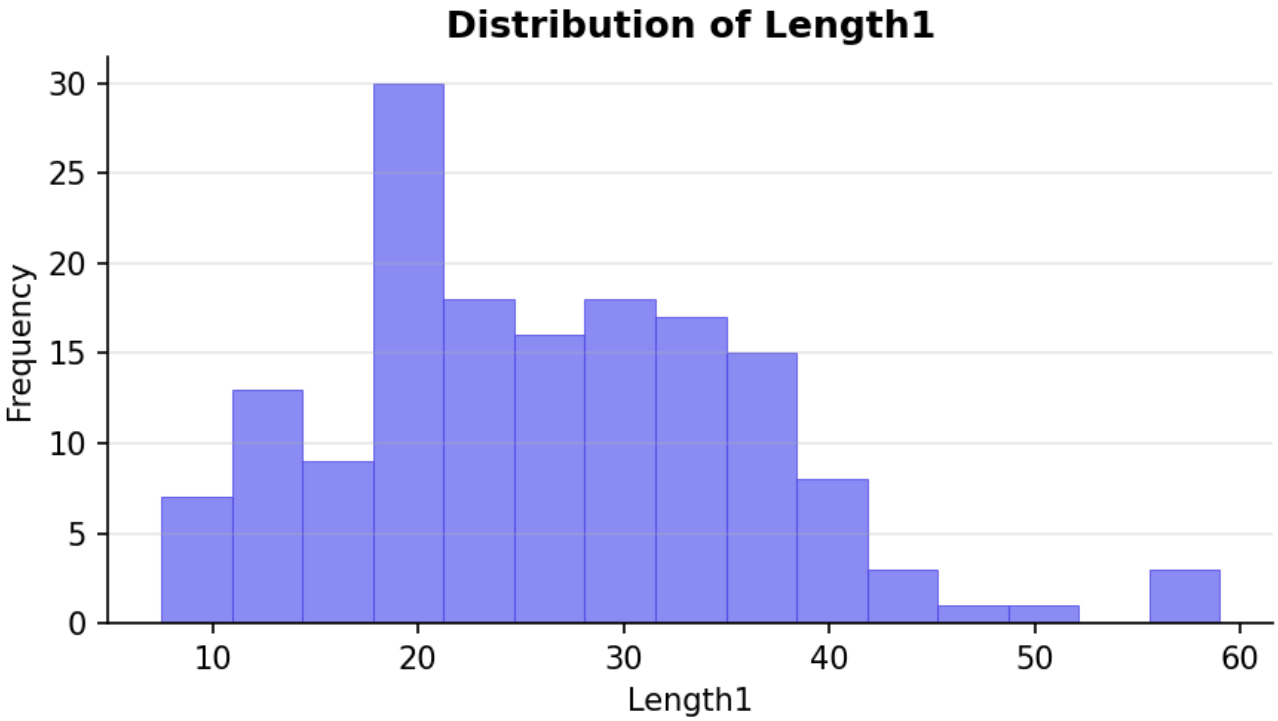
Outlier Analysis

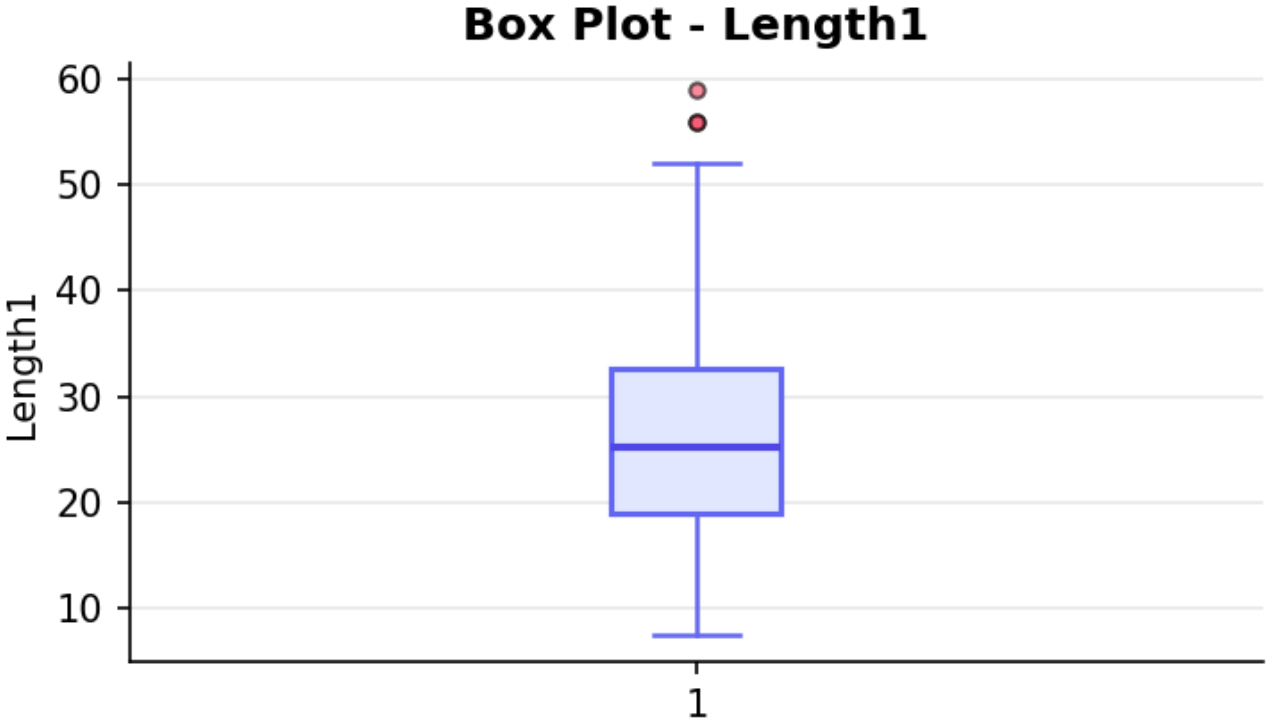
Column	Outliers	%	Q1	Q3	IQR	Lower Bound	Upper Bound
Weight	3	1.9%	120.00	650.00	530.00	-675.00	1445.00
Length1	3	1.9%	19.05	32.70	13.65	-1.43	53.18
Length2	3	1.9%	21.00	35.50	14.50	-0.75	57.25
Length3	1	0.6%	23.15	39.65	16.50	-1.60	64.40
Height	0	0.0%	5.94	12.37	6.42	-3.69	22.00
Width	0	0.0%	3.39	5.58	2.20	0.09	8.88

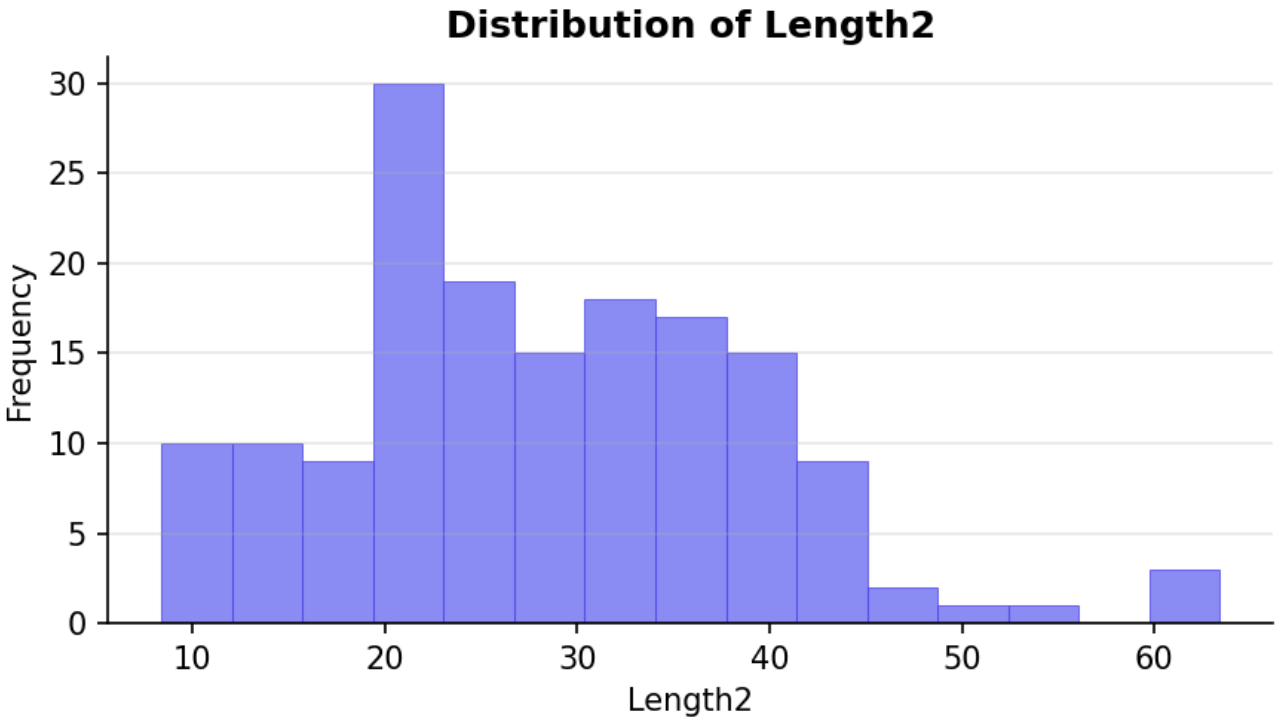
Distribution Analysis

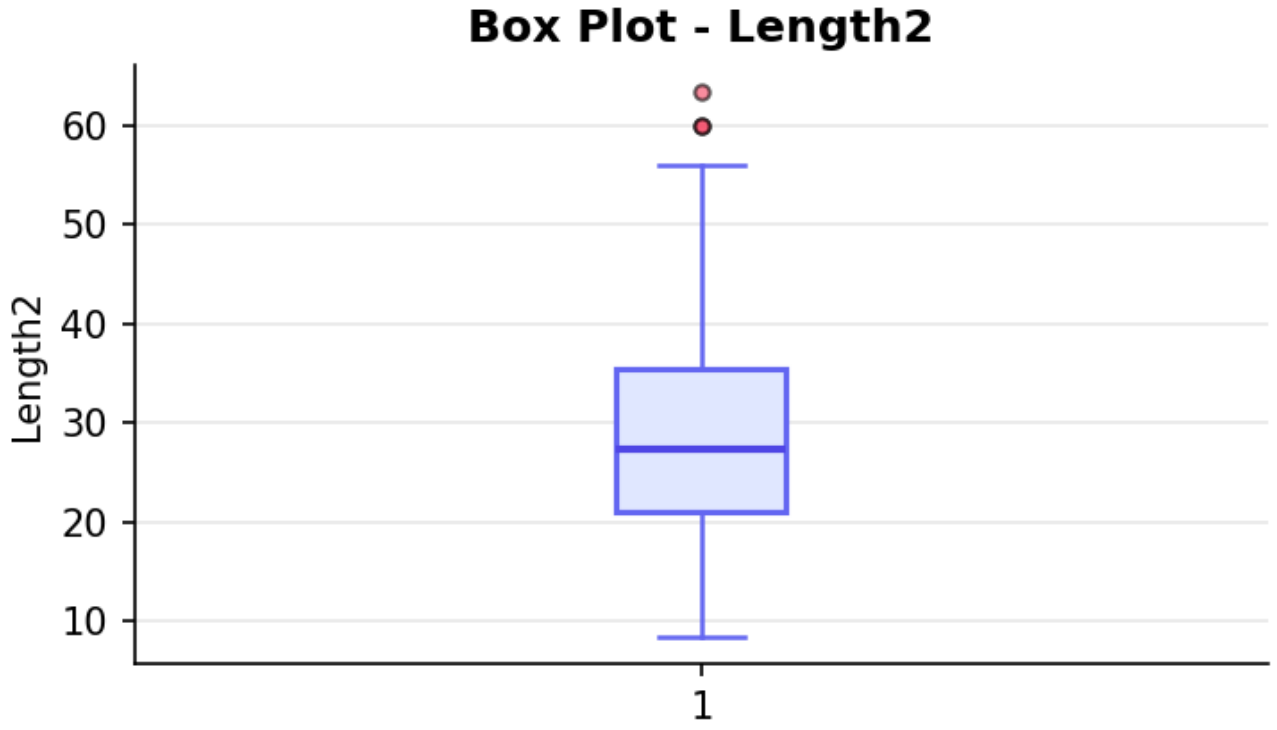


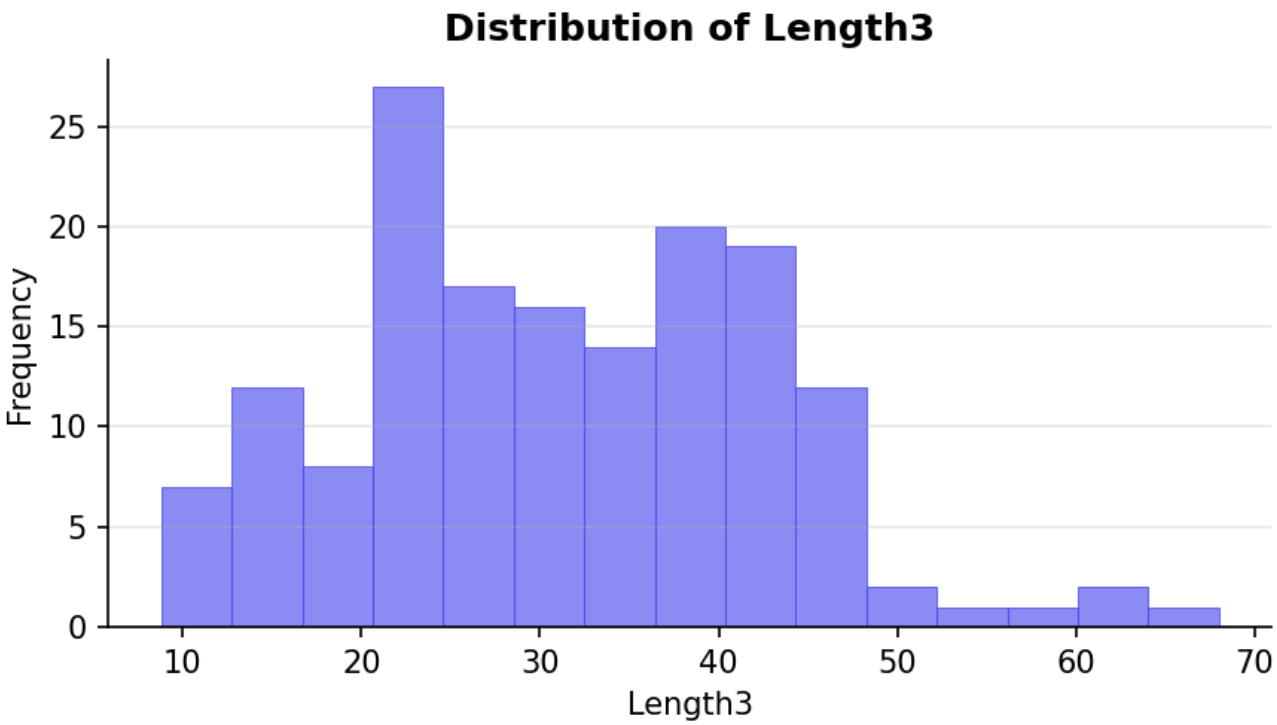


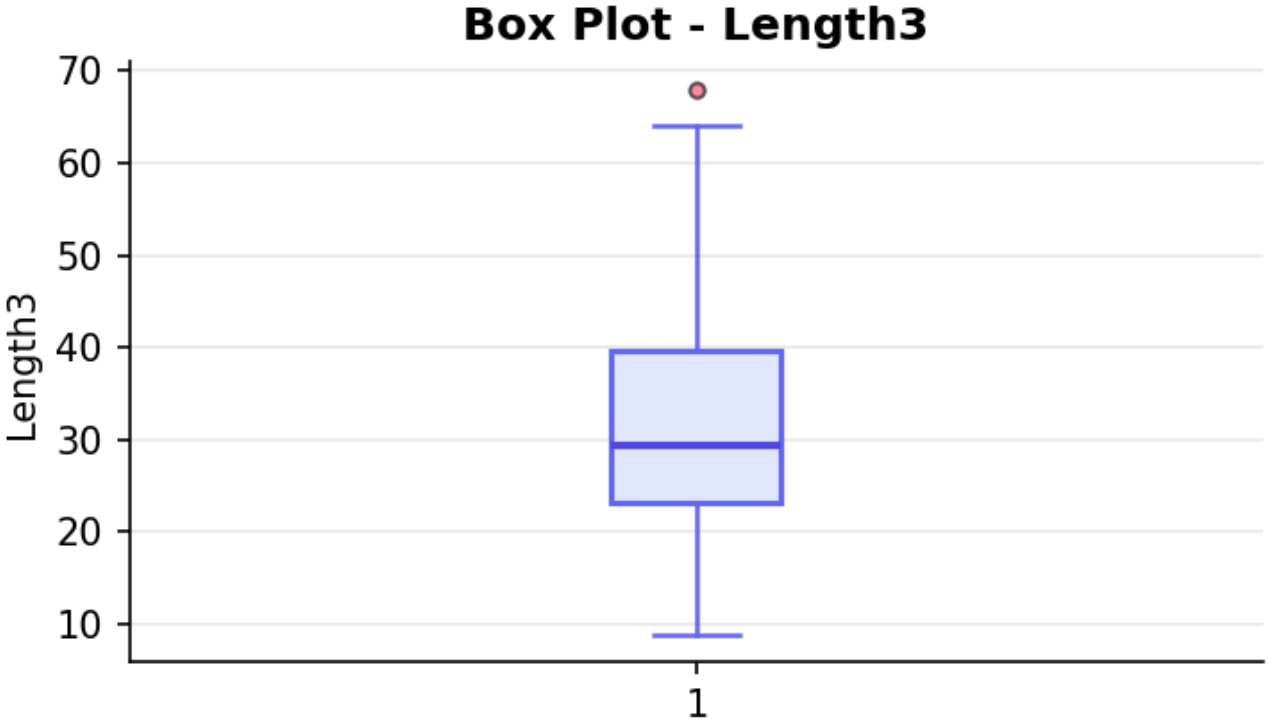

Distribution of Length1
Frequency
30
25
20
15
10
5
0
10
20
30
40
50
60
Length1

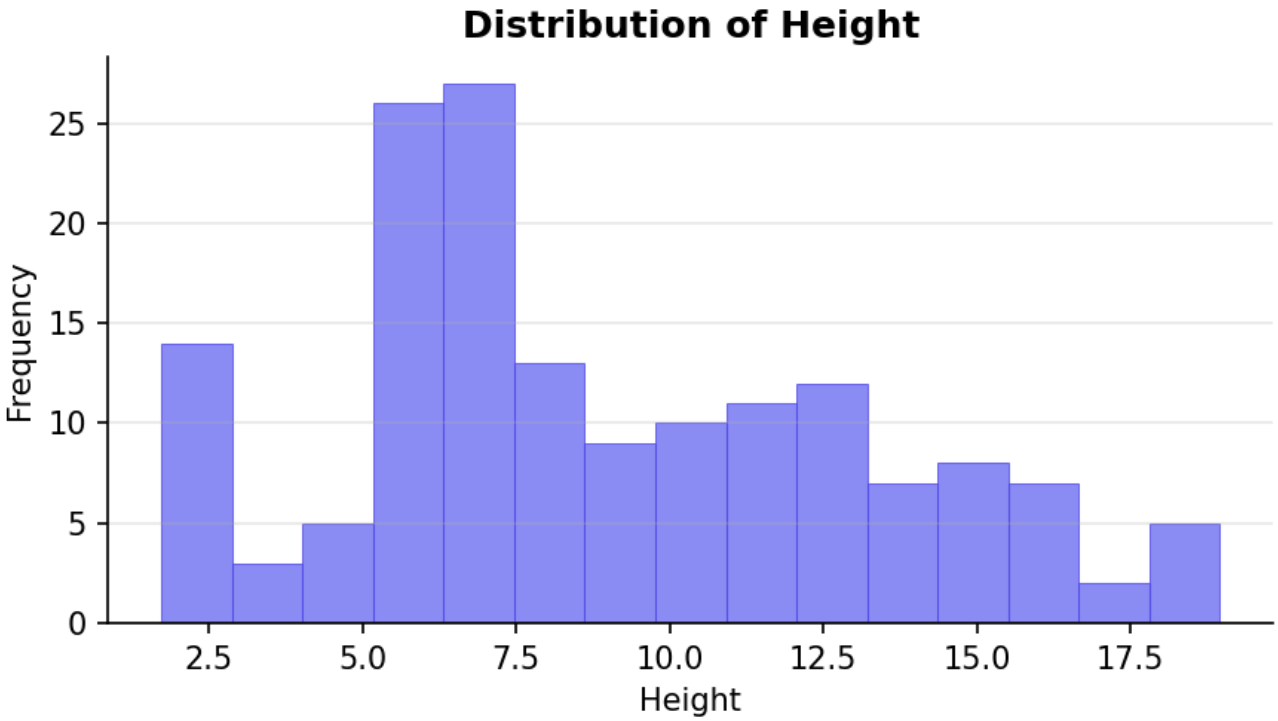

Box Plot - Length1
Length1
60
50
40
30
20
10
1

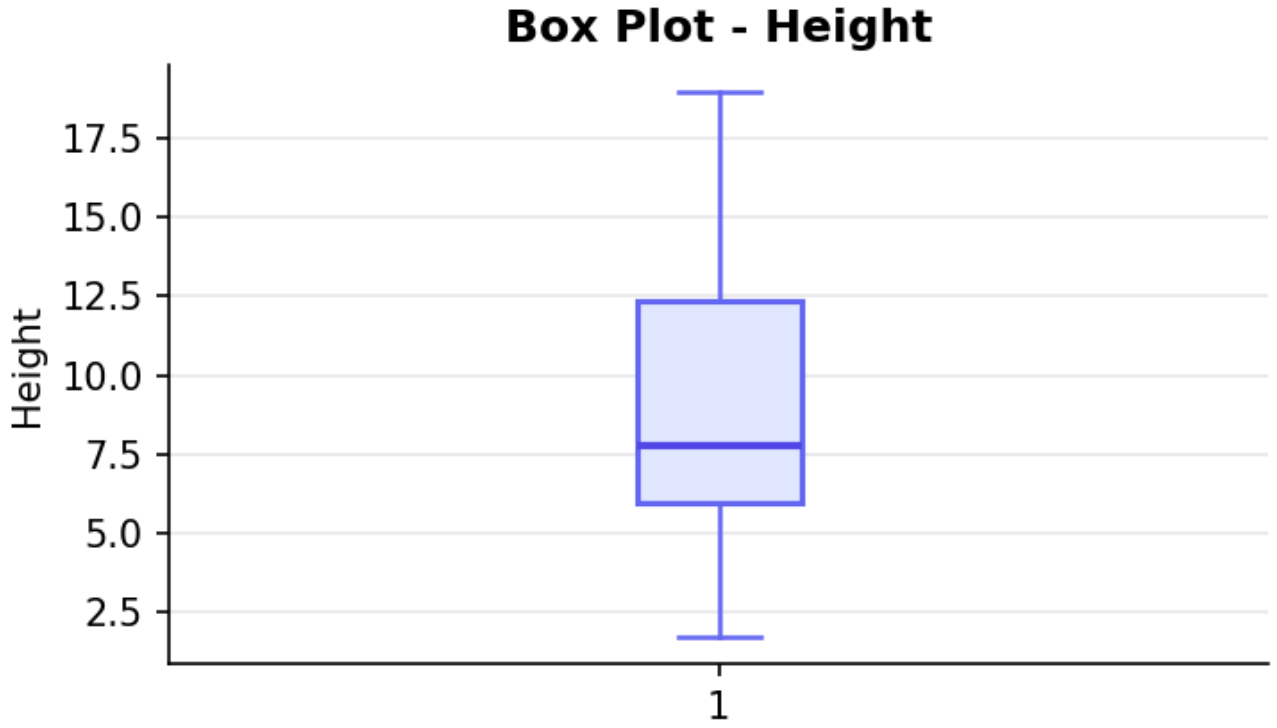

Distribution of Length2
Frequency
30
25
20
15
10
5
0
10
20
30
40
50
60
Length2

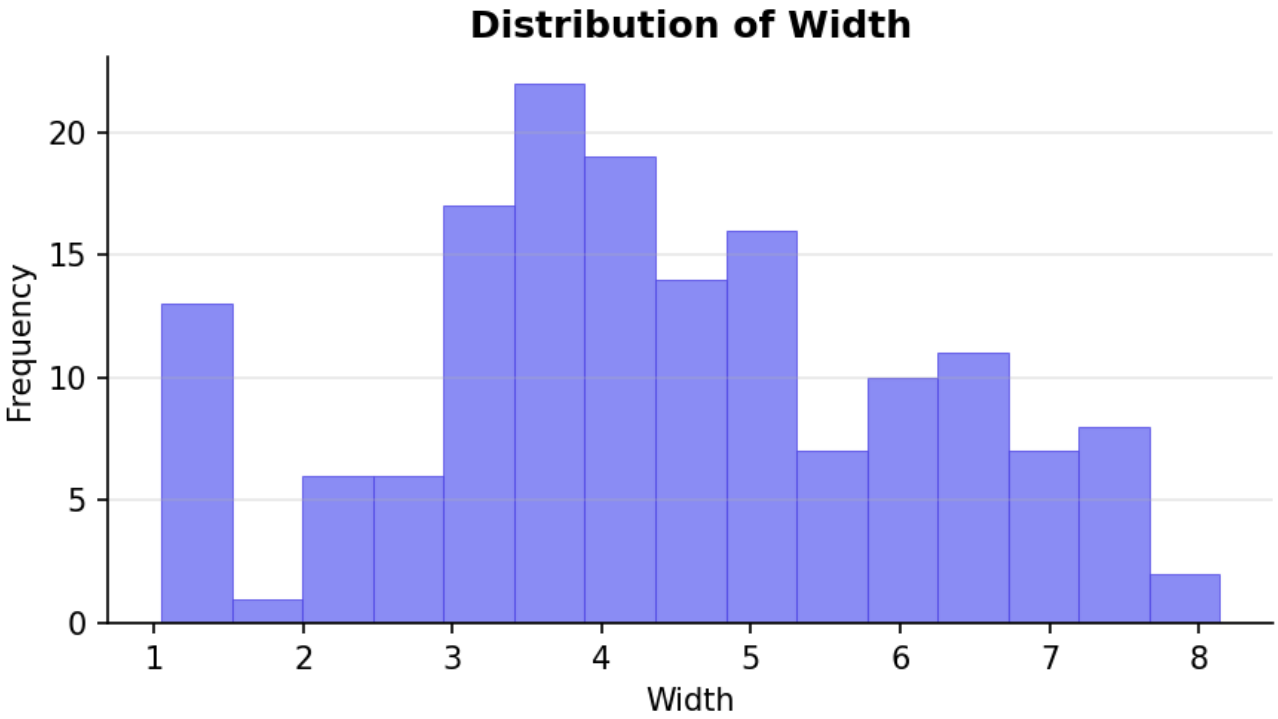

Box Plot - Length2
Length2
60
50
40
30
20
10
1

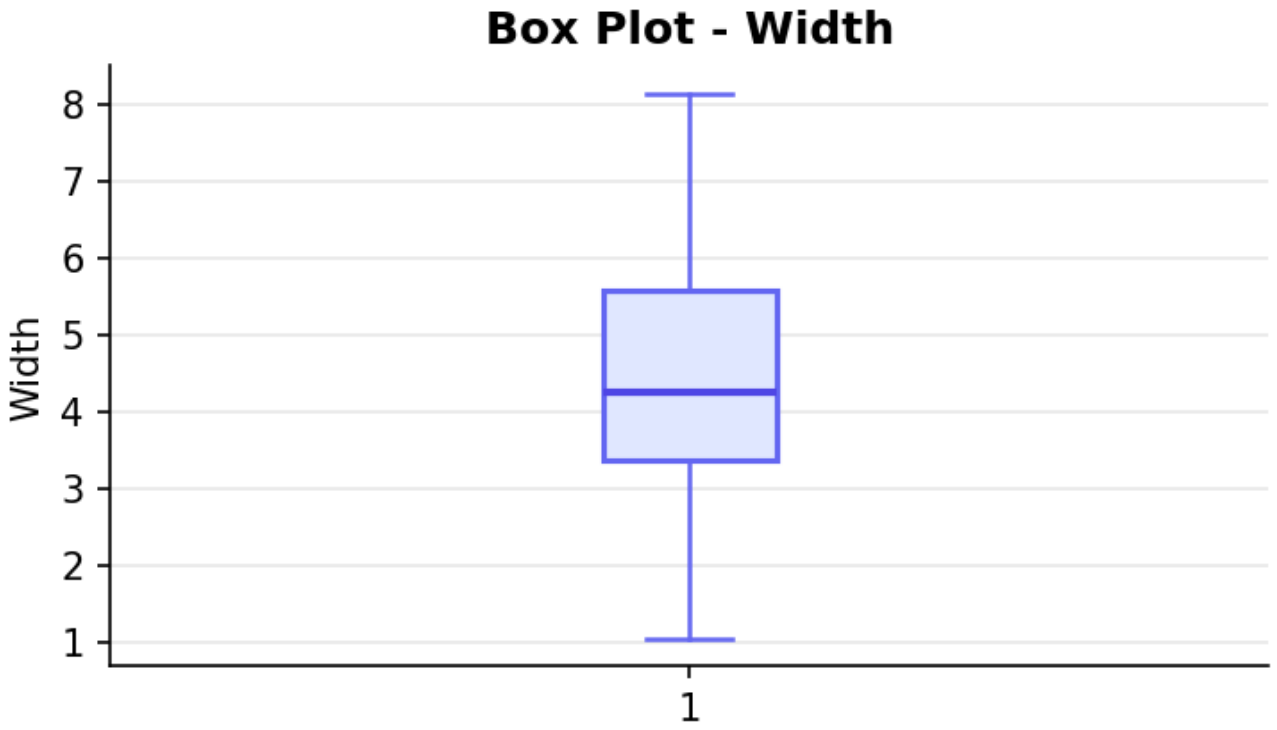

Distribution of Length3
Frequency
25
20
15
10
5
0
10
20
30
40
50
60
70
Length3


Box Plot - Length3
Length3
70
60
50
40
30
20
10
1


Distribution of Height
Frequency
0
5
10
15
20
25
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Height


Box Plot - Height
Height
2.5
5.0
7.5
10.0
12.5
15.0
17.5
1


Distribution of Width
Frequency
Width
0
5
10
15
20
1
2
3
4
5
6
7
8


Box Plot - Width
Width
1
2
3
4
5
6
7
8
1